

## DBMS – SELLER AND INVENTORY MANAGEMENT

Project Title: EcoCart – Sustainable Products E-Commerce Order Management System

Skill Task: DBMS

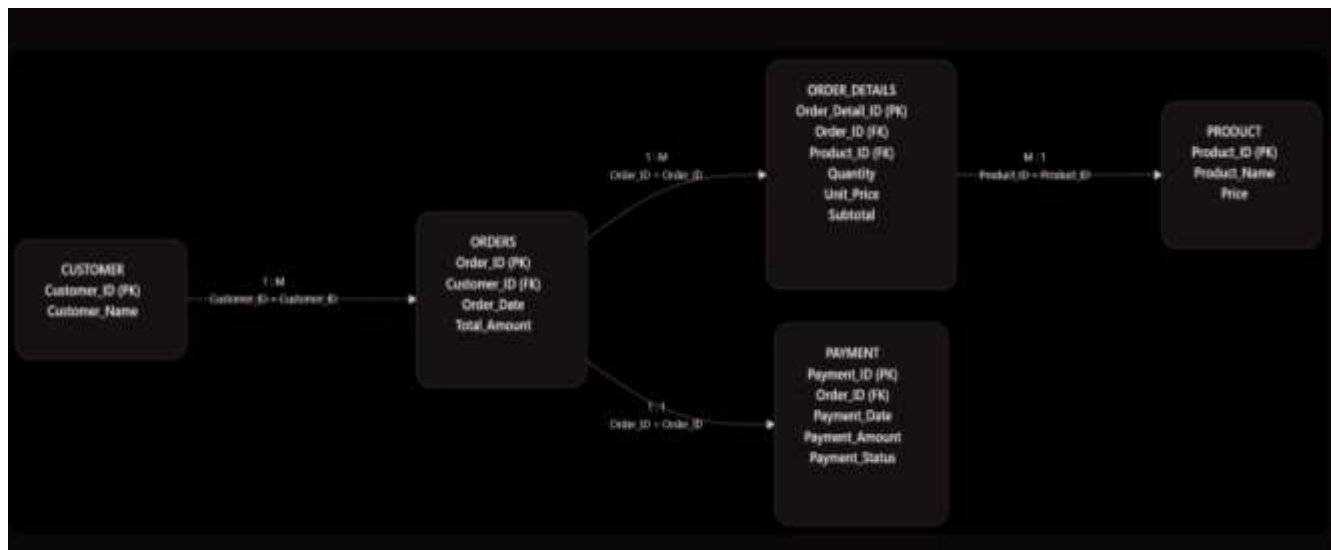
Week: 5

### 1. Database Design

#### 1.1 Payment Table

| Attribute             | Data Type    | Constraint  |
|-----------------------|--------------|-------------|
| <i>Payment_ID</i>     | INT          | Primary Key |
| <i>Order_ID</i>       | INT          | Foreign Key |
| <i>Payment_Mode</i>   | VARCHAR2(20) | NOT NULL    |
| <i>Payment_Date</i>   | DATE         | NOT NULL    |
| <i>Payment_Amount</i> | NUMBER(10,2) | NOT NULL    |
| <i>Payment_Status</i> | VARCHAR2(20) | NOT NULL    |

### 2. Table Relationships



### 3. Table Creation

#### 3.1 Create Payment Table

```
SQL> CREATE TABLE Payment (  
2     Payment_ID      NUMBER(5) PRIMARY KEY,  
3     Order_ID        NUMBER(5),  
4     Payment_Mode     VARCHAR2(20) NOT NULL,  
5     Payment_Date     DATE NOT NULL,  
6     Payment_Amount   NUMBER(10,2) NOT NULL,  
7     Payment_Status   VARCHAR2(20) NOT NULL,  
8     CONSTRAINT fk_payment_order  
9     FOREIGN KEY (Order_ID)  
10    REFERENCES Orders(Order_ID)  
11 );
```

Table created.

### 4. Payment Data Insertion

```
SQL> INSERT INTO Payment  
2 (Payment_ID, Order_ID, Payment_Mode, Payment_Date, Payment_Amount, Payment_Status)  
3 VALUES  
4 (501, 201, 'UPI', SYSDATE, 2500.00, 'Successful');
```

1 row created.

```
SQL> INSERT INTO Payment  
2 (Payment_ID, Order_ID, Payment_Mode, Payment_Date, Payment_Amount, Payment_Status)  
3 VALUES  
4 (502, 202, 'Card', SYSDATE, 1800.00, 'Successful');
```

1 row created.

```
SQL> INSERT INTO Payment  
2 (Payment_ID, Order_ID, Payment_Mode, Payment_Date, Payment_Amount, Payment_Status)  
3 VALUES  
4 (503, 203, 'Net Banking', SYSDATE, 3200.00, 'Failed');
```

1 row created.

```
SQL> INSERT INTO Payment  
2 (Payment_ID, Order_ID, Payment_Mode, Payment_Date, Payment_Amount, Payment_Status)  
3 VALUES  
4 (504, 204, 'UPI', SYSDATE, 1500.00, 'Successful');
```

1 row created.

## 5. Manage Payment Transactions

### 5.1 Display Successful Payments

```
SQL> SELECT *
      2 FROM Payment
      3 WHERE Payment_Status = 'Successful';
```

| PAYMENT_ID | ORDER_ID | PAYMENT_MODE | PAYMENT_D | PAYMENT_AMOUNT |
|------------|----------|--------------|-----------|----------------|
| 501        | 201      | UPI          | 16-SEP-26 | 2500           |
| 502        | 202      | Card         | 16-SEP-26 | 1800           |
| 504        | 204      | UPI          | 16-SEP-26 | 1500           |

### 5.2 Display Failed Payments

```
SQL> SELECT *
      2 FROM Payment
      3 WHERE Payment_Status = 'Failed';
```

| PAYMENT_ID | ORDER_ID | PAYMENT_MODE | PAYMENT_D | PAYMENT_AMOUNT |
|------------|----------|--------------|-----------|----------------|
| 503        | 203      | Net Banking  | 16-SEP-26 | 3200           |

## 6. Modify Payment Status

```
SQL> UPDATE Payment
  2 SET Payment_Status = 'Successful'
  3 WHERE Payment_ID = 503;
```

1 row updated.

```
SQL> SELECT *
  2 FROM Payment
  3 WHERE Payment_ID = 503;
```

| PAYMENT_ID | ORDER_ID | PAYMENT_MODE | PAYMENT_D | PAYMENT_AMOUNT |
|------------|----------|--------------|-----------|----------------|
| 503        | 203      | Net Banking  | 16-SEP-26 | 3200           |
| Successful |          |              |           |                |

## 7. Analyze Payment Methods

```
SQL> SELECT
  2 Payment_Mode,
  3 COUNT(*) AS Total_Transactions
  4 FROM Payment
  5 GROUP BY Payment_Mode;
```

| PAYMENT_MODE | TOTAL_TRANSACTIONS |
|--------------|--------------------|
| Net Banking  | 1                  |
| Card         | 1                  |
| UPI          | 2                  |

Calculate amount collected by payment mode

```
SQL> SELECT
  2 Payment_Mode,
  3 SUM(Payment_Amount) AS Total_Amount
  4 FROM Payment
  5 WHERE Payment_Status = 'Successful'
  6 GROUP BY Payment_Mode;
```

| PAYMENT_MODE | TOTAL_AMOUNT |
|--------------|--------------|
| Net Banking  | 3200         |
| Card         | 1800         |
| UPI          | 4000         |

## 8. Payment Transaction Report

```
SQL> SELECT
  2     p.Payment_ID,
  3     o.Order_ID,
  4     c.Customer_ID,
  5     c.Customer_Name,
  6     p.Payment_Mode,
  7     p.Payment_Date,
  8     p.Payment_Amount,
  9     p.Payment_Status
 10 FROM Payment p
 11 JOIN Orders o
 12     ON p.Order_ID = o.Order_ID
 13 JOIN Customer c
 14     ON o.Customer_ID = c.Customer_ID
 15 ORDER BY p.Payment_Date DESC;
```

| PAYMENT_ID | ORDER_ID | CUSTOMER_ID | CUSTOMER_NAME | PAYMENT_MODE | PAYMENT_D | PAYMENT_AMOUNT | PAYMENT_STATUS |
|------------|----------|-------------|---------------|--------------|-----------|----------------|----------------|
| 504        | 204      | 104         | Karthik S     | UPI          | 16-SEP-26 | 1500           | Successful     |
| 503        | 203      | 103         | Priya         | Net Banking  | 16-SEP-26 | 3200           | Successful     |

| PAYMENT_ID | ORDER_ID | CUSTOMER_ID | CUSTOMER_NAME | PAYMENT_MODE | PAYMENT_D | PAYMENT_AMOUNT | PAYMENT_STATUS |
|------------|----------|-------------|---------------|--------------|-----------|----------------|----------------|
| 502        | 202      | 102         | Rahul         | Card         | 16-SEP-26 | 1800           | Successful     |
| 501        | 201      | 101         | Anitha Kumar  |              |           |                |                |

| PAYMENT_ID | ORDER_ID | CUSTOMER_ID | CUSTOMER_NAME | PAYMENT_MODE | PAYMENT_D | PAYMENT_AMOUNT | PAYMENT_STATUS |
|------------|----------|-------------|---------------|--------------|-----------|----------------|----------------|
|            |          |             |               | UPI          | 16-SEP-26 | 2500           | Successful     |